
Generic Online Message Rest Based Specifications

Version 1.5

1. Revision History

Date	Version	Authored By	Revision Section	Description
March 2021	1.0	Faizan Ahmed Siddiqui	Initial Draft	
March 2022	1.1	Ovais Tahir / Faizan Ahmed Siddiqui	Input parameters	Reserved field length updated Due date format is updated in response packet Bill status section updated in Response string
April 08,2022	1.2	Faizan Ahmed Siddiqui	Input parameters	Bundle id added
April 14, 2022	1.3	Faizan Ahmed Siddiqui	Input parameters	Fetch Bundle method is added.
May 9, 2022	1.4	Faizan Ahmed Siddiqui	Input parameters	Bundle id length updated Fetch bundle response code added
Apr 18, 2023	1.5	Faizan Ahmed Siddiqui	Response Coee	Inquiry Response Code Mapping Added



Rest Message Specifications

2. Introduction

Document consists of following two transactions, which are implemented through web service.

- 1 Fetch Bundle
- 2 Inquiry
- 3 Payment

1) Fetch Bundle

Input Parameters

URL: http://ip:port/v1/Transaction/Fetchbundle

Method Type: POST

Request:

Parameter	Description	Mandatory/Optional	Data Type
		Payment (U/B/P)	
PCID	Company UCID	M	String(8) Company UCID Required

```
{  
  "PCID": "MBLINK01"  
}
```

Response:

Parameter	Description	Mandatory/Optional	Data Type
companyId	Company UCID	M	String (N) 8
Response Code	Response code	M	String (N)2
BillerName	This will contain Biller Name	M	String (A) 30
bundleDetails	"bundleId": "1212", "bundleName": "Weekly Plus", "description": "abc Weekly Plus", "expiryDate": "27-MAR-22", "amount": "230", tag": PrePaid, 7days, 'Onnet min' '30' ; 'ofnet min ' '60'; 15min (1 am to 9pm), any data	M	List
bundleDetails	"bundleId": "1678", "bundleName": "Weekly Plus", "description": "abc Weekly Plus", "expiryDate": "27-MAR-22", "amount": "230", tag": "PrePaid, 7days, 'Onnet min' '30' ; 'ofnet min ' '60'; 15min (1 am to 9pm), any data"	M	List
bundleDetails	etc.	M	List

Note: tag max (2000)length will be use to pass below fields

- Bundle Category,
- Validity,
- Bundle Detail,
- Addition info,
- Reserved



Rest Message Specifications

```
{
  "companyId": "MBLINK01",

  "responseCode": "00",

  "billerName": "Mobilink APC Prepaid",
  "bundleDetails": [
    {
      "bundleId": "1212",
      "bundleName": "Weekly Plus",
      "description": "abc Weekly Plus",
      "expiryDate": "27-MAR-22",
      "amount": "230",
      "tag": "PrePaid, 7days, 'Onnet min' '30' ; 'ofnet min' '60'; 15min (1 am to 9pm), any data"
    },
    {
      "bundleId": "1678",
      "bundleName": "SuperCard",
      "description": "abc Super Card",
      "expiryDate": "27-MAR-22",
      "amount": "480",
      "tag": "PrePaid, 7days, 'Onnet min' '30' ; 'ofnet min' '60'; 15min (1 am to 9pm), any data"
    }
  ]
}
```

1) Inquiry

This transaction is required to check the validity of the transaction. The Details of the function are as under:

Input Parameters

Parameter	Description	Mandatory/Optional	Data Type
		Payment (U/B/P)	
username		M	String Maximum length (60), in case length is less than 60 then there will no whitespaces padded
password		M	String Maximum length (60), in case length is less than 60 then there will no whitespaces padded
consumer_number	This will be either consumer number	M	String maximum length (24) In case of consumer number field length less than 24 then there will be no whitespace padded.

			**At 1LINK it maximum string length is (20)
bank_mnemonic	This field contains either Bank Mnemonic or UC Id. Banks' Mnemonic e.g. BoK, QNB etc UCId e.g. SSGC0001, UFONE003 etc	M	For Bank Mnemonic string(8)
reserved	Reserved field	O	String (400)

Inquiry Reserved field Details		
Parameter	Description	Data Type
Reserved		CNIC(13) Account Id(28) BundleID(100) Supporting Information 1(100) Supporting Information 2(144) In case of empty or less length, fields will padded with space as all above fields are fixed length

Request

URL : ../api/1.0/Payments/BillInquiry

Method Type: POST

Headers:



Rest Message Specifications

Header["username"] = "Test"

Header["password"] = "@bcd"

Body:

```
{
  "consumer_number": "14211101052009",
  "bank_mnemonic": "KESC0001",
  "reserved": "something, special, string, can, be, send, into, it."
}
```

Response

For Success Case:

Http Status: 200

```
{
  "response_Code": "00",
  "consumer_Detail": "MUHAMMAD FEROUZ",
  "bill_status": "U",
  "due_date": "20210329",
  "amount_within_dueDate": "-0000000186900",
  "amount_after_dueDate": "+0000000202500",
  "billing_month": "0809",
  "date_paid": "20180301",
  "amount_paid": "000000202500",
  "tran_auth_id": "202500",
  "reserved": "something, special, string, can, be, send, into, it."
}
```

For Error Case:

Http Status: 400, 404, 500 etc.

Return String in following format

Parameter	Description	Mandatory/Optional	Data Type
response_Code	'00' in case of valid consumer number that exists in the system/database of company and status is active. '01' in case of invalid consumer number that does not exist. '02' in case of valid consumer number that is currently blocked or dormant/inactive (i.e. transactions are not allowed). '03' Unknown Error/Bad Transaction. '04' Invalid Data '05' Processing Failed	M	String (N) 2
consumer_detail	This will contain Consumer Name	M	String (A) 30 Left justified right padded with spaces
bill_status	'U' in case of unpaid bill or bill on which payment is allowed 'P' in case of paid bill on which further payment is not allowed 'B' in case of blocked bill Partial (Multiple/ Excess) payment companies will always send 'T' because excess payments can be done	M	String (A) 1
due_date	Due Date will be provided ..	M	yyyyMMdd

amount_within_duedate	Amount will be provided	M	AN14 - Last 2 digits minor units - Left padded with zeros e.g. Amount120.00Rs is formatted as +0000000012000 Amount-120.00Rs is formatted as -0000000012000
amount_after_duedate	Amount will be provided	M	AN14 - Last 2 digits minor units - Left padded with zeros e.g. Amount 120.00 Rs is formatted as +0000000012000 Amount-120.00Rs is formatted as -0000000012000
billing_month	BillingMonth willbeprovided	M	String N(4) yyMM
date_paid	In case bill status is 'P' else white spaceshouldbesent according to specified length	C	String N(8) yyyyMMdd
amount_paid	In case bill status is 'P' else white spaces should be sent according to the specified length	C	String N(12) - Last 2 digits minor units - Left padded with zeros e.g. Amount 120.00 Rs is formatted as 000000012000
tran_auth_id	In case billstatus is 'P' else white spaceshouldbesendaccording to specified length	C	String N(6) e.g. 698243
reserved	Reserved field	0	String (400)



Rest Message Specifications

Inquiry Reserved field Details		
Parameter	Description	Data Type
Reserved		Total loan amount (C) (12) Total outstanding amount (C)(12) Fee Amount (C) (12) Max Limit (C) (12) min Limit (C) (12) Supporting Information 1(C) (200) Supporting Information 2(C) (67) In case of empty or less length, fields will padded with space as all above fields are fixed length

2) Payment Transaction

Input Parameter

Parameter	Description	Mandatory/Optional Data Type	
		Payment (U/B/P/T)	
username		M	String Maximum length (60), in case length is less than 60 then there will no whitespaces padded
password		M	String Maximum length (60), in case length is less than 60 then there will no whitespaces padded
consumer_number (PK)	This will be consumer no	M	String maximum length (24) In case of consumer number field length less than 24 then there will be no whitespace padded. **At 1LINK it maximum string length is (20)
tran_auth_id (PK)	Transaction Authorization ID	M	String N(6) e.g. 698243



Rest Message Specifications

Transaction_amount	Credit Amount	M	String AN12 - Last 2 digits minor units - Left padded with zeros e.g.Amount120.00 Rs is formatted as 000000012000
tran_date (PK)	Transaction Date	M	String N(8) YYYYMMDD
tran_time (PK)	Transaction Time	M	String N(6) HHMMSS



Rest Message Specifications

bank_mnemonic	This field contains either Bank Mnemonic or UCId. Usually. Banks' Mnemonic e.g. BoK, QNB etc UCID e.g. SSGC0001, Ufone003 etc	M	ForBankMnemonic string(8) ForUCID string(8)
Reserved	Reserved field		String (400)

Payment Reserved field Details		
Parameter	Description	Data Type
Reserved		Payer CNIC (13) City (30) Province (20) Account Id (28) fromAccountType 2 N(X) fromAccountTitle 30 AN(X) BundleID (100) Supporting Information 1 (100) Supporting Information 2 (192) In case of empty or less length, fields will padded with space as all above fields are fixed length

Request

URL : ../api/1.0/Payments/BillPayment

Method Type: POST

Headers:

Header["username"] = "Test"

Header["password"] = "@bcd"

Body:

```
{
  "consumer_number": "14211101052009",
  "tran_auth_id" : "112233",
  "transaction_amount": "000000012000"
  "tran_date" : "20180911",
  "tran_time" : "121366",
  "bank_mnemonic" : "KESC0001",
  "reserved": "something, special, string, can, be, send, into, it."
}
```

Response

For Success Case:

Http Status: 200

```
{
  "response_Code": "00",
  "Identification_parameter" : "MUHAMMAD FERROZ",
  "reserved": "something, special, string, can, be, send, into, it."
}
```

For Error Case:

Http Status: 400, 404, 500 etc

Return String in following format

Parameter	Description	Mandatory/Optional		Data Type

response_Code	00 is success 01 Mobilenumber not found 02 unknown error 03 Duplicate Transaction 04 Invalid Data 05 ProcessingFailed Note: more response codes may be added where ever applicable			M	N(2)
Identification_parameter	Voucher Number, Scratch Card Number etc			O	String Length: (20)
reserved	Reserved field - first 20 digits will be the serial number			O	String (400)

Payment Reserved field Details		
Parameter	Description	Data Type
Reserved		Supporting Information 1 (200) Supporting Information 2 (200) In case of empty or less length, fields will padded with space as all above fields are fixed length

Response Code

TransactionType	Response Code	Response Code Description
Inquiry	00	Successful Bill Inquiry
Inquiry	01	Consumer Number does not exist
Inquiry	02	Consumer Number Block
Inquiry	03	Unknown Error / Bad Transaction
Inquiry	04	Invalid Data (e.guserid, password, Bank Mnemonic provided wrong)
Inquiry	05	Service Fail
Inquiry	06	Bill Already Paid
Payment	00	Successful Bill Payment
Payment	01	Consumer Number does not exist
Payment	02	Unknown Error / Bad Transaction
Payment	03	Duplicate Transaction
Payment	04	Invalid Data (e.guserid, password, Bank Mnemonic provided wrong)
Payment	05	Service Fail
Payment	06	Bill Already Paid
Fetch bundle	00	Successful inquiry
Fetch bundle	01	Consumer Number does not exist
Fetch bundle	04	Invalid Data
Fetch bundle	05	Service Fail

Duplication Transaction

It is possible that Web Service will receive same payment transaction multiple times (duplicate transaction) so it is necessary that Web Service should able to handle duplicate transaction. A bill payment transaction can be identified as unique by using following fields.

1. Consumer_Number
2. Tran_Auth_Id



Rest Message Specifications

3. Tran_Date
4. Tran_Time

Http Response Codes

Status	Description
200	Approve Response Code
400	Bad Request/Invalid Information



Rest Message Specifications

401	Unauthorized
403	Forbidden
404	Not found
500	Internal Server Error